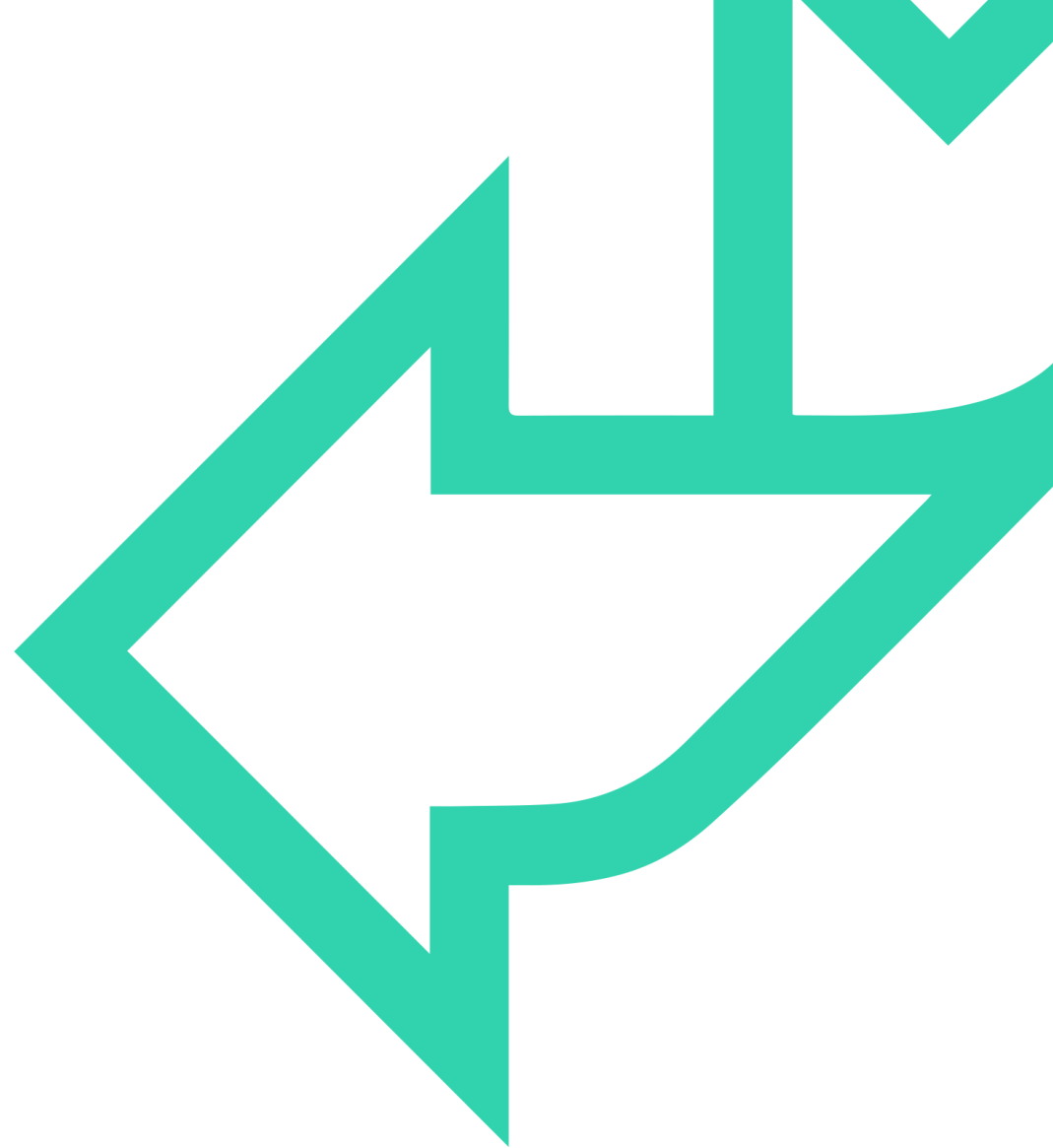




Lists





LESSON OBJECTIVES

In this chapter, you will learn how to:

- Create lists
- Read an item from a list
- Add an item to a list
- Remove an item from a list
- Replace an item





PYTHON LISTS



List of elements in memory

```
numbers = [1,3,5,7]
```

1
3
5
7

```
names = ['Bob', 'Steve', 'Helen']
```

Bob
Steve
Helen



ADDRESS AN ELEMENT USING ITS INDEX



```
numbers = [1,3,5,7,9]
```

```
print(numbers[0])  
print(numbers[2])
```

[0]	1
[1]	3
[2]	5
[3]	7

```
names = ["Bob", "Steve", "Helen"]
```

```
print(names[1])
```





THE LEN() FUNCTION

Use the `len()` function to get the length of a list

```
numbers = [1,3,5,7]
```

```
print(len(numbers))
```

4

```
names = ["Bob", "Steve",  
"Helen"]
```

```
print(len(names))
```

3



ITERATING THROUGH A LIST - FOR LOOP



```
names = ["Bob", "Steve",  
         "Helen"]
```

```
for name in names:  
    print(name)
```

```
Bob  
Steve  
Helen
```

```
Press any key to continue ...
```



ITERATING THROUGH A LIST – WHILE LOOP

```
names = ["Bob", "Steve", "Helen"]
```

```
i=0      # used as index
```

```
while(i < len(names) ):  
    print(names[i])  
    i += 1
```

```
Bob  
Steve  
Helen
```

```
Press any key to continue ...
```



STRINGS (LISTS OF CHARACTERS)



```
greeting = "Hello"  
for x in greeting:  
    print(x)
```

```
H  
e  
l  
l  
o  
Press any key to continue ...
```

```
greeting = "Hello"  
i=0      # used as index  
  
while(i < len(greeting)):  
    print(greeting[i])  
    i += 1
```




APPENDING ELEMENTS AT THE END

```
numbers = [1,3,5,7,9]
```

```
numbers.append(88)
```

```
[1, 3, 5, 7, 9, 88]
```

You can start with an empty List:

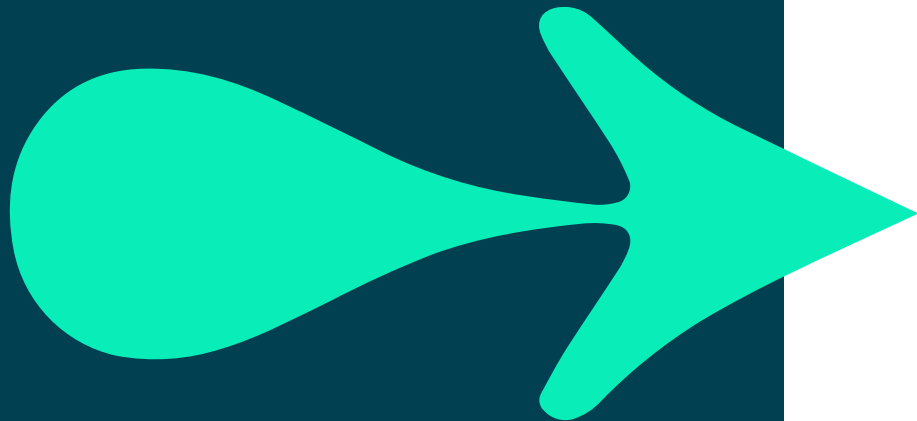
```
nos = []
```

```
nos.append(10)
```

```
nos.append(20)
```

```
nos.append(30)
```

```
[10, 20, 30]
```



REMOVING ELEMENTS

```
numbers = [1,3,5,7,5,9,5]
```

```
numbers.remove(5)
```

```
[1, 3, 7, 5, 9, 5]
```

```
del(numbers[0])
```

```
[3, 7, 5, 9, 5]
```

```
names = ['Bob', 'Steve', 'Helen']
```

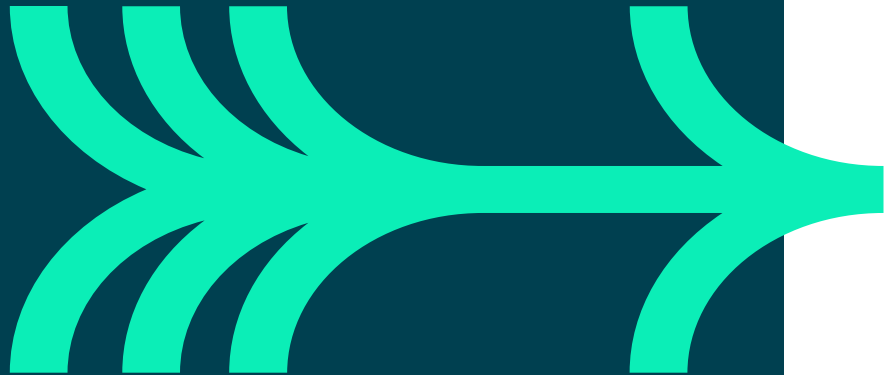
```
names.remove("Steve")
```

```
['Bob', 'Helen']
```

Works with any kind of
List



CHECKING FOR EXISTENCE



Use the **in** command to check if an item is present in a list

```
adminIDs = [12,33,84,45,67,36,16,66,67,99]
id = int(input('Enter your ID '))
if id in adminIDs:
    print('Welcome!')
```

numeric

```
names = ['David','John','Joanne','Sean','Sonia']
if 'Sean' in names:
    print('Sean is in the list!')
```

strings

```
names = "David,John,Joanne,Sean,Sonia"
if 'John' in names:
    print('John is in the list!')
```

List of characters
(string)



SORTING LIST ELEMENTS



Use the `sort()` function to sort elements

```
ages = [12,33,84,45,67,36,16]
```

```
ages.sort()  
print(ages)
```

[12, 16, 33, 36, 45, 67, 84]

```
ages = [12,33,84,45,67,36,16]
```

```
ages.sort(reverse=True)  
print(ages)
```

[84, 67, 45, 36, 33, 16, 12]



THE STRING.SPLIT() FUNCTION

Used for splitting and extracting elements from a string using a delimiter

```
data = 'Bob,Steve,Helen'  
names = data.split(',')  
print(names)
```

['Bob', 'Steve', 'Helen']

```
data='18/OCT/2020'  
parts = data.split('/')  
print(parts)
```

['18', 'OCT', '2020']



PYTHON FUNDAMENTALS



In this lesson, you've learnt how to:

- Create a list
- Read an item from a list
- Add an item to a list
- Remove an item from a list
- Replace an item



GUIDED TASK

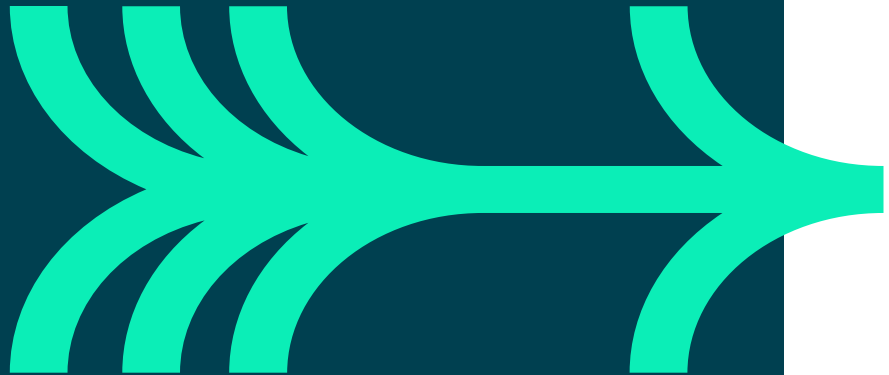
→ Please open the instructions in Cloud Academy

- This guided task has many exercises
- **See how many you can do!**



FURTHER READING

- <https://www.python.org/>
- <https://www.python.org/dev/peps/pep-0008/#a-foolish-consistency-is-the-hobgoblin-of-little-minds>





Thank you!

Hope you enjoyed this learning journey!